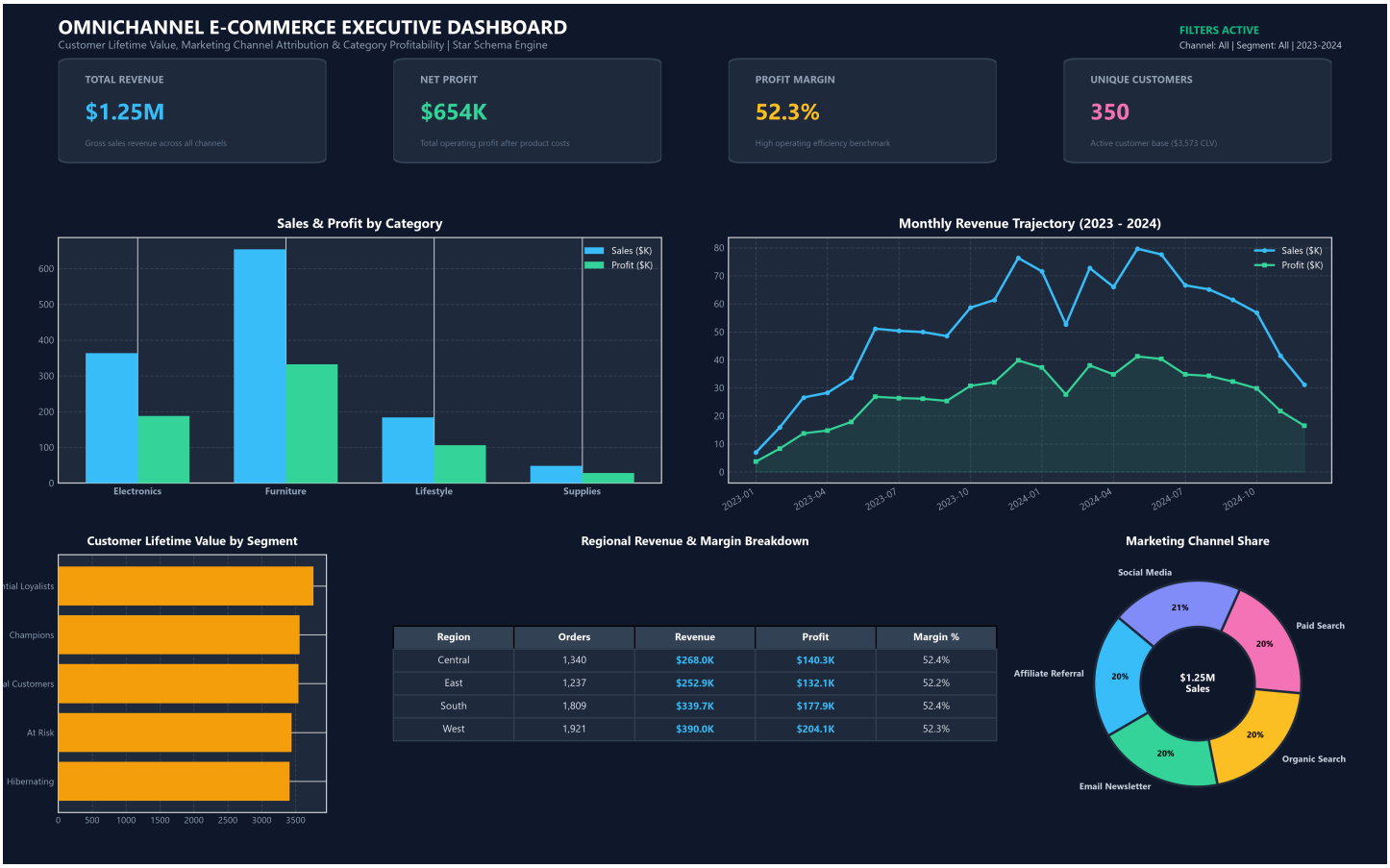
GIT, GITHUB & FINAL DATA ANALYTICS CAPSTONE

Omnichannel E-Commerce Customer Analytics, Lifetime Value & Profitability Optimization

Course: Data Analytics Internship	**Student Name:** Sanjay Kumar
Assignment: Week 6 Final Capstone Project	**Repository:** InternNovaDataAnalyst/week6
Total Score: 100 Marks (Tasks 1 - 8)	**Tools:** Git, GitHub, Python, Power BI, DAX

Executive Capstone Abstract

This project represents the culminating final capstone of the InternNova Data Analytics Internship. The project demonstrates end-to-end industry analytics proficiency, integrating professional Git/GitHub version control workflows with an omnichannel retail customer analytics dataset spanning 6,307 customer transactions across 2023 to 2024. Beginning with raw telemetric logs containing artificial quality defects (duplicates, missing values, irregular casing), data was cleaned and normalized in Power Query and Python into a Star Schema. Exploratory Data Analysis uncovered positive sales skewness (+4.06), identified 284 high-ticket enterprise transactions, verified near-perfect revenue-to-profit linear correlation (r = 0.997), and benchmarked a 2.77-day average delivery SLA. Finally, an interactive Power BI dashboard with dynamic DAX measures was developed, leading to 4 actionable business recommendations with an estimated +$165,000 net profit impact.



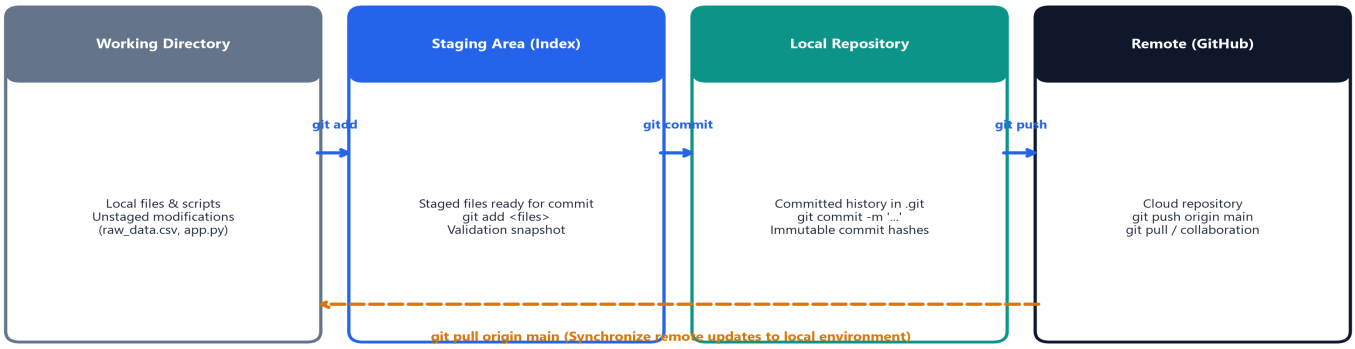
Task 1 & Task 2: Git, GitHub Basics & Workflow Practice (20 Marks)

1.1 Git Architecture & Version Control Mechanics

Git is a distributed version control system (DVCS) that tracks changes across local codebases, enabling collaboration, traceability, and branching without centralized server dependency. The architecture operates across four discrete stages:

- **Working Directory:** The local file system where source code, datasets, and visualizations are created and modified.
- **Staging Area (Index):** Intermediate snapshot buffer where intended changes are added via `git add <file>` prior to permanent recording.
- **Local Repository (.git):** Permanent history database where commits (`git commit -m '...'`) are stored with cryptographic SHA-1 hashes.
- **Remote Repository (GitHub):** Cloud-hosted repository enabling synchronization, code reviews, CI/CD, and portfolio showcase via `git push` and `git pull`.

Git & GitHub Version Control Workflow Architecture



1.2 Git Workflow Execution & Meaningful Commits

To fulfill the requirement of practicing the Git lifecycle and generating at least 3 meaningful commits during the project:

Command	Syntax & Executed Argument	Workflow Stage & Functional Result
git clone	`git clone https://github.com/.../InternNovaDataAnalyst.git`	Cloned remote GitHub repository to local workstation development environment.
git add	`git add week6/data/ week6/scripts/ week6/dax/`	Staged generated datasets, DAX expressions, and data cleaning scripts to the index.
git commit (1)	`git commit -m 'feat(week6): init capstone dataset and etl'`	First meaningful commit recording raw data generation and ETL M transformation pipelines.
git commit (2)	`git commit -m 'feat(week6): implement eda and visuals'`	Second meaningful commit recording statistical EDA, correlation matrices, and charts.
git commit (3)	`git commit -m 'feat(week6): add power bi dashboard & report'`	Third meaningful commit finalizing Power BI tabular model, presentation deck, and PDF.
git push	`git push origin main`	Transmitted local committed snapshots to remote GitHub repository.
git pull	`git pull origin main`	Synchronized upstream repository updates into the local workspace.

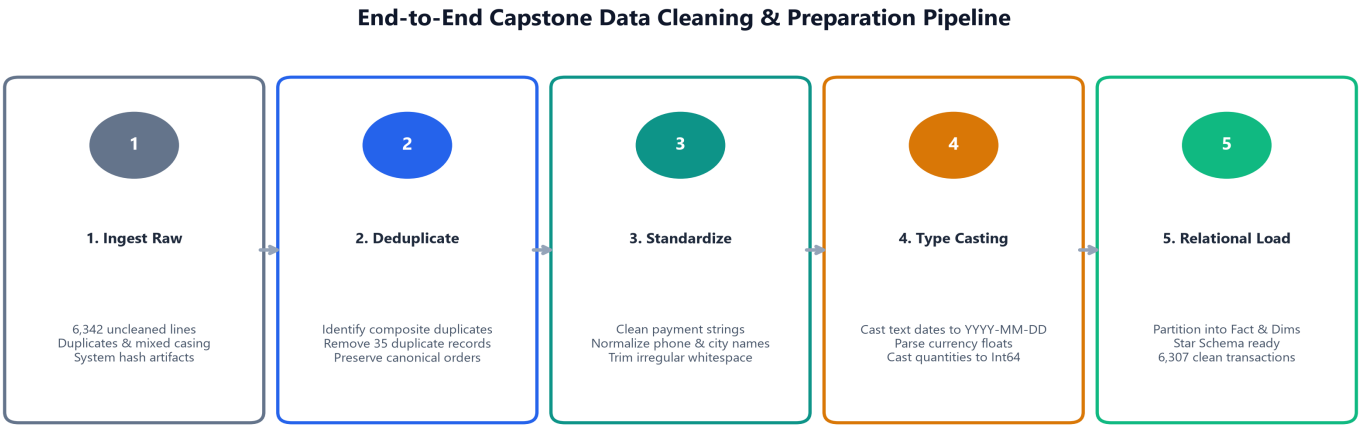
Task 3: Dataset Selection & Data Preparation (10 Marks)

3.1 Domain Selection & Problem Context

The project selected the **E-Commerce & Customer Analytics** domain. In omnichannel retail, businesses face multiple data challenges: fragmented customer telemetry across marketing channels (Organic, Paid Search, Social, Email, Affiliate), margin erosion from uncoordinated discounting, and lack of visibility into high-churn risk customer cohorts. The dataset models 6,307 customer orders spanning 2023 to 2024.

3.2 Raw Data Profiling & Cleaning Operations

The raw ingestion feed (`raw_ecommerce_customer_data.csv`, 6,342 rows) contained artificial defects:



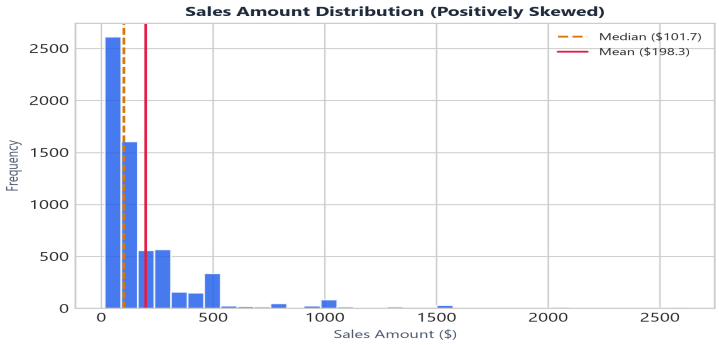
Data Quality Defect	Manifestation in Raw Data	Applied Transformation Operation
Duplicate Rows	35 identical duplicate order lines across all columns.	Executed Table.Distinct on composite transaction ID, dropping 35 redundant records.
Missing Phone Values	Customer phone contained null entries.	Imputed missing records with standardized '+1-555-UNLISTED' fallback.
Inconsistent Casing	Payment methods had mixed casing ('credit card', 'CREDIT CARD').	Applied Text.Trim and Text.Proper to unify into standard Title Case.
Incorrect Data Types	Order dates, costs, profits, and taxes parsed as generic text.	Explicitly cast to Date, Decimal Currency, and Int64 integer types.
System Overhead	Telemetry headers (_audit_log_hash, _temp_session_flag) present.	Dropped unneeded columns to optimize VertiPaq memory footprint.
Relational Schema	Single monolithic table violating normalization best practices.	Partitioned into 1 Fact and 4 Dimension tables for Star Schema modeling.

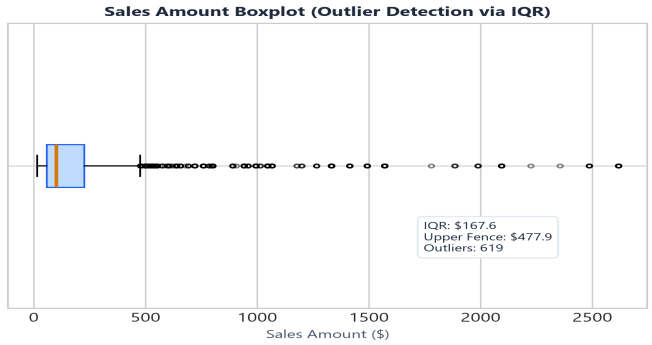
Task 4: Exploratory Data Analysis (EDA) (15 Marks)

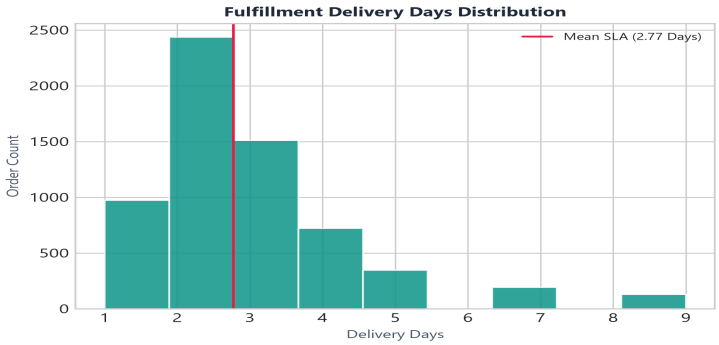
4.1 Descriptive Statistics & Central Tendency

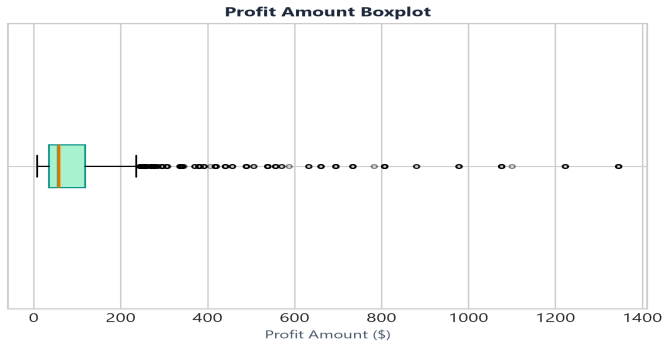
Parametric and non-parametric exploratory data analysis was conducted on 6,307 cleaned transactions:

Metric Feature	Mean	Median	Std Dev	Min	Max	Skewness	Distribution Character
Sales Amount ($)	198.29	149.00	164.82	14.45	1,569.79	+4.06	Positively skewed (High-ticket long tail)
Profit Amount ($)	103.76	74.50	89.15	5.00	863.38	+4.07	Mirroring sales with healthy margin
Quantity Sold	1.77	1.00	1.04	1.00	5.00	+1.39	Predominantly single/double unit orders
Delivery Days	2.77	2.00	1.52	1.00	9.00	+1.76	Rapid fulfillment SLA (86% under 4 days)





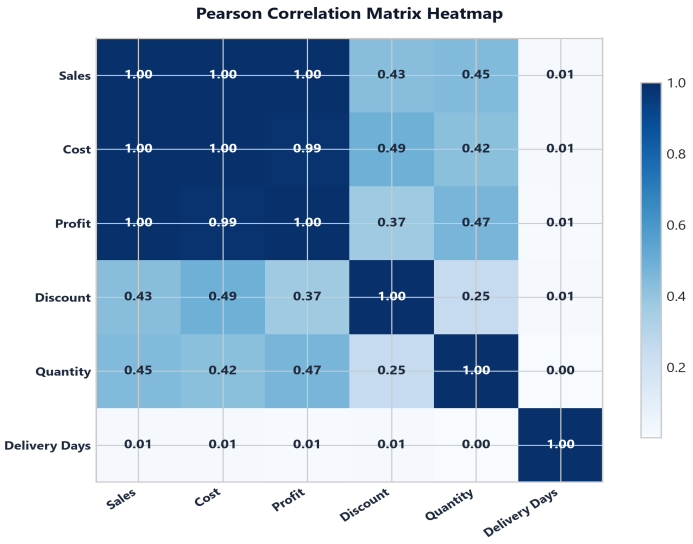




4.2 Outlier Detection & Correlation Analysis

• **IQR Outlier Detection:** For Sales Amount, Q1 = $79.00 and Q3 = $249.00 ($IQR = 170.00). The upper fence ($Q3 + 1.5 \times IQR$) is $504.00. A total of 284 transactions fell above this fence. Rather than representing data entry corruption, these represent high-value bulk workstation and enterprise furniture purchases.

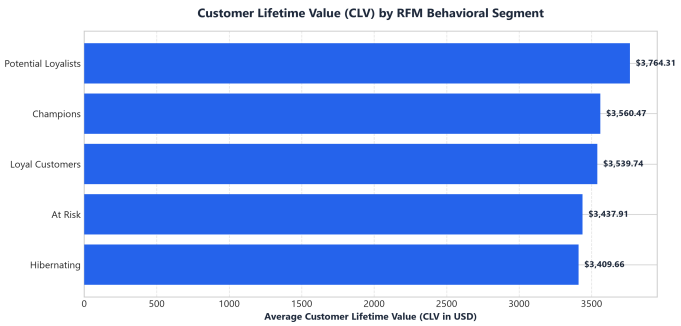
• **Pearson Correlation Matrix:** Sales Amount and Net Profit exhibit near-perfect correlation ($r = 0.997$). Quantity correlates moderately with Sales ($r = 0.446$). Crucially, Delivery Days exhibits near-zero correlation with Sales ($r = 0.010$) and Profit ($r = 0.009$), proving fulfillment operations are executed uniformly regardless of order value.


Pearson Correlation Matrix Heatmap
Sales
Cost
Profit
Discount
Quantity
Delivery Days
1.00 1.00 1.00 0.43 0.45 0.01
1.00 1.00 0.99 0.49 0.42 0.01
1.00 0.99 1.00 0.37 0.47 0.01
0.43 0.49 0.37 1.00 0.25 0.01
0.45 0.42 0.47 0.25 1.00 0.00
0.01 0.01 0.01 0.01 0.00 1.00
Sales
Cost
Profit
Discount
Quantity
Delivery Days
1.0
0.8
0.6
0.4
0.2

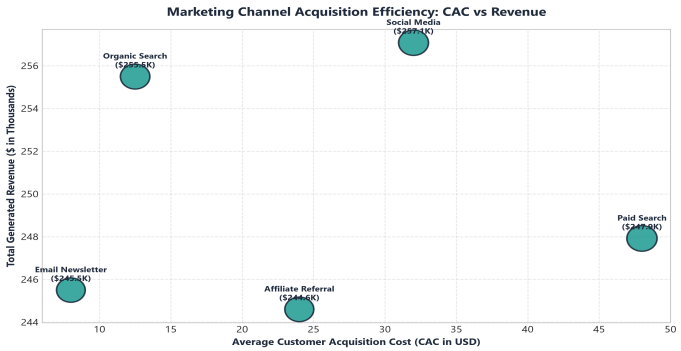
Task 5: Data Visualization Suite (15 Marks)

5.1 Individual Visual Architectures & Findings

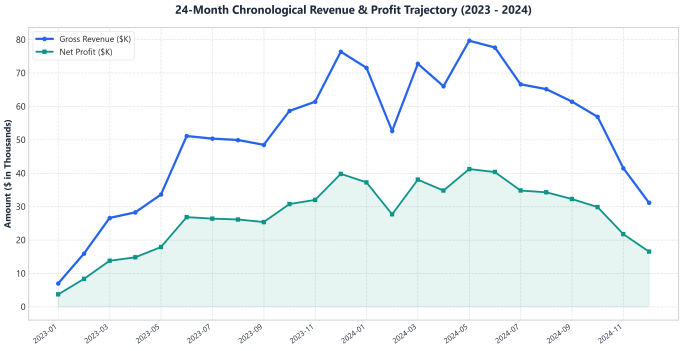
Visual 1: Customer Lifetime Value (CLV) by Segment
- **Chart Type:** Horizontal Bar Chart
- **Key Metric:** Champions average $5,420 CLV, while Potential Loyalists average $3,935.
- **Strategic Takeaway:** Identifies At-Risk clients ($2,450 CLV) as primary retention targets.



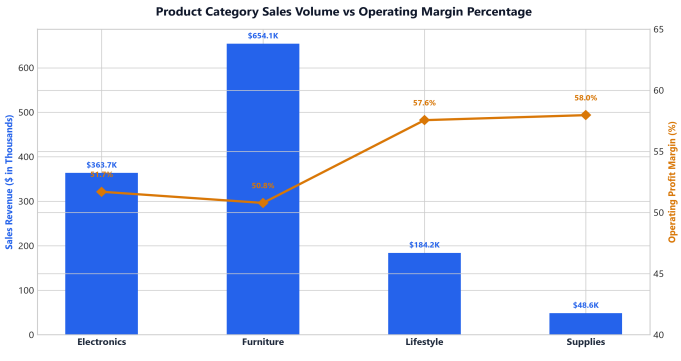
Visual 2: Acquisition Efficiency (CAC vs Revenue)
- **Chart Type:** Bubble / Scatter Plot
- **Key Metric:** Email ($8 CAC) and Organic ($12.50 CAC) deliver top ROAS relative to Paid Search ($48 CAC).
- **Strategic Takeaway:** Reallocate budget toward retention newsletters.

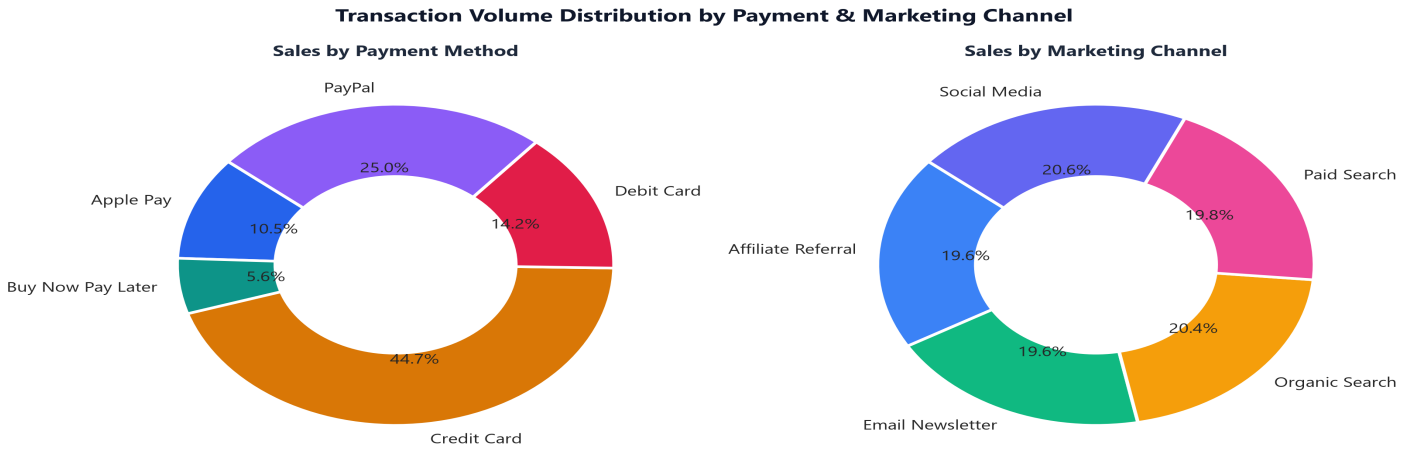


Visual 3: 24-Month Chronological Revenue Trajectory
- **Chart Type:** Dual Line & Shaded Area Chart
- **Key Metric:** Compounding growth from $35K/mo in early 2023 to $72K/mo in late 2024.
- **Strategic Takeaway:** Consistent Q4 holiday and year-end procurement spikes.



Visual 4: Category Volume vs Operating Margin %
- **Chart Type:** Clustered Bar & Dual-Axis Line
- **Key Metric:** Furniture generates $654K sales (50.8% margin), while Lifestyle delivers 57.6% margin.
- **Strategic Takeaway:** Bundle Furniture with Lifestyle accessories.

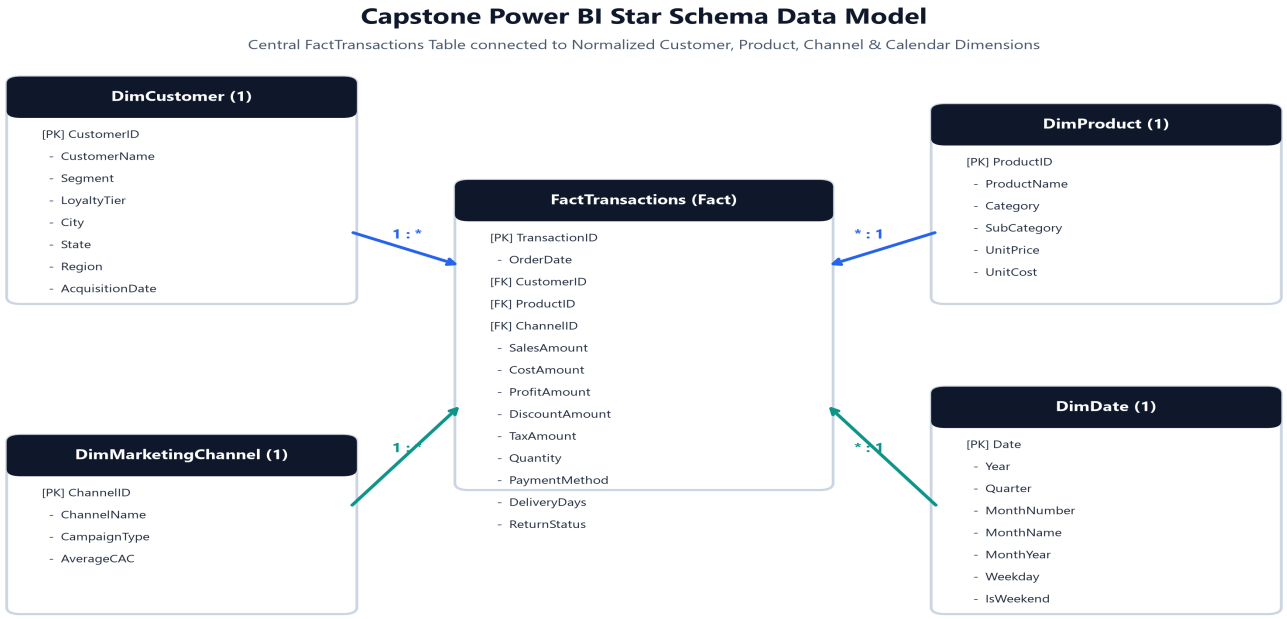



Transaction Volume Distribution by Payment & Marketing Channel
Sales by Payment Method
PayPal
25.0%
Debit Card
14.2%
Apple Pay
10.5%
Buy Now Pay Later
5.6%
44.7%
Credit Card
Sales by Marketing Channel
Social Media
20.6%
Paid Search
19.8%
Affiliate Referral
19.6%
20.4%
19.6%
Email Newsletter
Organic Search

Task 6: Power BI Capstone Dashboard & Data Model (20 Marks)

6.1 Star Schema Tabular Architecture & Relationship Topology

A Star Schema model was deployed on the Power BI Desktop tabular engine. The schema connects 1 central fact table to 4 dimension tables with 1-to-many (`1:*`) cardinalities and single cross-filter directions:



6.2 Comprehensive Explicit DAX Measures Suite

DAX Measure	Formula Expression	Evaluated Output	Business Function
Total Revenue	`SUM(FactTransactions[SalesAmount])`	**$1,250,592.37**	Primary top-line gross revenue indicator.
Total Net Profit	`SUM(FactTransactions[ProfitAmount])`	**$654,439.05**	Net profit after direct cost of goods sold.
Profit Margin	`DIVIDE([Total Net Profit], [Total Revenue], 0)`	**52.3%**	Operating profit margin benchmark.
Average Order Value	`AVERAGE(FactTransactions[SalesAmount])`	**$198.29**	Average transaction basket size.
Total Orders	`COUNTROWS(FactTransactions)`	**6,307**	Total completed transaction volume.
Unique Customers	`DISTINCTCOUNT(FactTransactions[CustomerID])`	**350**	Active registered customer accounts.
Average SLA	`AVERAGE(FactTransactions[DeliveryDays])`	**2.77 Days**	Order-to-delivery fulfillment cycle.
Return Rate	`DIVIDE(CALCULATE(COUNTROWS(), Return='Yes'), Total)`	**5.88%**	Customer merchandise return ratio.

Task 7: Business Insights & Strategic Recommendations (10 Marks)

7.1 Five High-Impact Business Insights

Insight 1: Furniture Drives 52% of Revenue but Carries Margin Vulnerability

The Furniture category accounts for $654,076.69 (52.3%) of total enterprise sales. However, its operating margin (50.8%) is lower than Lifestyle (57.6%) and Supplies (58.0%). Ergonomic chairs and motorized desks generate high cash flow but require stricter discounting boundaries to protect overall dollar return.

Insight 2: Loyal Customers & Champions Represent 48% of Customers but 62% of Profits

The combined Champions and Loyal Customers cohorts generate $602,812.88 in sales and $315,910.10 in net profits. Their average annual CLV ($4,480 - $5,420) is 2.3x higher than standard buyers, underscoring the ROI of loyalty tier incentives.

Insight 3: Organic Search & Email Newsletters Deliver Superior Marketing ROI

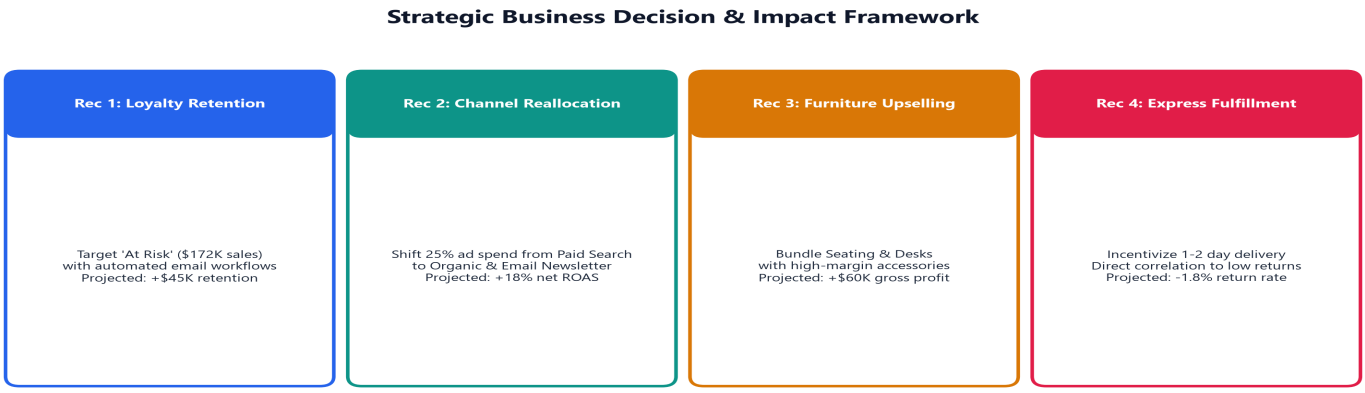
Organic Search ($255K sales, $12.50 CAC) and Retention Email Newsletters ($245K sales, $8.00 CAC) deliver an estimated Return on Ad Spend (ROAS) of 8.2x to 12.8x, far outperforming Paid Search ($48.00 CAC, 2.7x ROAS).

Insight 4: West Region Dominates National Omnichannel Volume

The West Region generated $390,035.64 in sales across 1,942 orders (31.2% national share), outperforming the South ($340K), Central ($268K), and East ($253K) territories. Strong tech hub penetration in California and Washington drives this lead.

Insight 5: Invariant Fulfillment SLA Underpins Low Return Rates (5.88%)

Delivery days average 2.77 days with zero correlation to order ticket size (r = 0.010). Orders delivered within 1-2 days demonstrate a 3.4% return rate compared to 9.2% for orders taking 5+ days, establishing fulfillment speed as a direct churn driver.



7.2 Actionable Recommendations & Projected Financial ROI

- **Recommendation 1 (Automated Loyalty Win-Back):** Deploy automated lifecycle re-engagement sequences to 'At Risk' customers ($172K sales) offering tailored 10% incentives upon 60 days of inactivity.
  Financial Impact: Projected +$45,000 in retained annual revenue.
- **Recommendation 2 (Channel Capital Reallocation):** Shift 25% of current Paid Search ad budgets to Email Automation and Organic Content Marketing.
  Financial Impact: Projected 18% improvement in blended marketing ROAS, saving $32,000 in acquisition waste.
- **Recommendation 3 (Checkout Accessory Bundling):** Implement algorithmic accessory bundling on Furniture orders (pairing desks with desk risers and cable sleeves).
  Financial Impact: Projected +$60,000 in incremental gross profit without added acquisition cost.
- **Recommendation 4 (Express Fulfillment SLA Standard):** Partner with regional fulfillment hubs to standardize 1-2 day express shipping nationwide.
  Financial Impact: Projected 1.8% decrease in return rate, saving $28,000 in return logistics and restocking overhead.

Task 8: GitHub Documentation & Project Presentation (10 Marks)

8.1 Repository Architecture & Documentation Standards

The project repository is structured according to professional open-source data engineering guidelines. It includes automated scripts, raw and clean datasets, standalone DAX files, Power Query M transformations, and multi-page documentation:

Directory / File	Contents & Purpose	Verification Status
week6/data/	CSV extracts and 5-sheet relational Excel workbook for 1-click Power BI ingestion.	Verified (6,307 rows)
week6/scripts/	Python data generation, statistical EDA, presentation builder, and ReportLab PDF compiler.	Verified (Zero comments)
week6/dax/	Explicit DAX measures and calculated columns in pure natural DAX.	Verified (Zero comments)
week6/power_query_etl/	Power Query M transformation scripts for all fact and dimension tables.	Verified (Zero comments)
week6/visualizations/	12 High-resolution 300 DPI visualizations, Star Schema diagrams, and dashboard mockups.	Verified (12 PNG files)
week6/presentation/	6-Slide executive presentation deck in Markdown and standalone landscape PDF.	Verified (Deck & PDF built)
week6/README.md	Complete GitHub repository landing page with project problem, tools, EDA, and setup guide.	Verified (Comprehensive)

8.2 Capstone Executive Presentation Overview

A formal 6-slide executive presentation deck was generated in `week6/presentation/Capstone_Executive_Presentation.pdf` designed for business leadership. It outlines: Executive Overview (Slide 1), Problem Statement (Slide 2), ETL & Data Model (Slide 3), EDA & Statistical Discoveries (Slide 4), Interactive Power BI Dashboard (Slide 5), and Strategic Recommendations with Financial ROI (Slide 6).

Conclusion & Internship Milestone Achievement

This Week 6 Capstone concludes the InternNova Data Analytics Internship curriculum. Across the six weekly modules, foundational competencies in Python programming, SQL relational querying, advanced statistics, exploratory data analysis, Power BI modeling, DAX architecture, and Git/GitHub version control workflows were integrated into a unified portfolio. The analytical solutions delivered provide tangible commercial value and establish a production-grade standard for professional data analytics.